

Physiology of the Kidneys

Human Physiology Chapter 17

Biology Teaching Resources

December 24, 2025

Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Gross Structure of the Urinary System

- The **urinary system** (泌尿系统) consists of kidneys, ureters, bladder, and urethra
- The kidneys filter blood and produce urine to regulate body fluid composition
- Ureters transport urine from kidneys to bladder through peristaltic contractions
- The bladder stores urine; urethra allows urine excretion



Gross Structure of the Urinary System

- Each kidney receives blood from the **renal artery** (肾动脉) and returns it via **renal vein** (肾静脉)
- Blood flow to kidneys is approximately 1100 ml/min, about 20% of cardiac output
- This high blood flow is essential for kidney's filtration and regulatory functions

Internal Kidney Structure

- The kidney has two main regions: **renal cortex** (肾皮质) and **renal medulla** (肾髓质)
- **Renal pyramids** (肾锥体) in the medulla contain collecting ducts
- Urine drains through **calyces** (肾盏) and **renal pelvis** (肾盂) into the ureter
- This anatomical organization is essential for concentrating urine



Internal Kidney Structure

- Imaging studies reveal the pathway of urine flow through the urinary system
- The calyces, renal pelvis, ureters, and bladder form a continuous drainage system
- Understanding this anatomy is crucial for diagnosing urinary tract disorders



Internal Kidney Structure

- The renal vasculature follows a hierarchical branching pattern
- Blood flows: renal artery → interlobar arteries → arcuate arteries → interlobular arteries
- This vascular arrangement ensures uniform blood supply to all nephrons
- Proper blood flow is essential for maintaining glomerular filtration



The Nephron: Functional Unit

- The **nephron** (肾单位) is the functional unit; each kidney has over 1 million nephrons
- Each nephron consists of a glomerulus, tubule segments, and associated blood vessels
- Blood is filtered at the glomerulus, and the filtrate is modified as it flows through tubules



The Nephron: Functional Unit

- Two types of nephrons serve different functions
- **Cortical nephrons** (85%) have short loops, mainly in the cortex
- **Juxtamedullary nephrons** (15%) have long loops extending into the medulla
- Juxtamedullary nephrons are crucial for producing concentrated urine



Control of Micturition

- **Micturition** (排尿) is the process of urination controlled by neural reflexes
- The **detrusor muscle** (逼尿肌) contracts to empty the bladder
- Two sphincters control urine release: internal (smooth muscle) and external (skeletal muscle)
- The **guarding reflex** (防护反射) prevents involuntary emptying
- The **voiding reflex** (排尿反射) is controlled by the pons and higher brain centers
- Voluntary control develops between ages 2-3

Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

The Filtration Barrier

- The glomerular capillaries are highly specialized for filtration
- **Podocytes** (足细胞) wrap around capillaries with foot processes
- Spaces between foot processes form **filtration slits** (滤过裂隙)
- This unique structure allows selective filtration based on size and charge



The Filtration Barrier

- The filtration barrier has three layers preventing protein passage
- Capillary endothelium has **fenestrae** (窗孔) for water and small solutes
- **Basement membrane** (基底膜) provides structural support
- **Slit diaphragm** (裂隙膜) between podocyte foot processes is the main barrier



The Filtration Barrier

- Electron microscopy reveals the detailed structure of the filtration barrier
- The three-layered barrier allows selective passage of molecules
- Large proteins are retained in the blood
- Damage to this barrier causes **proteinuria** (蛋白尿)



Glomerular Ultrafiltrate Formation

- **Glomerular ultrafiltrate** (肾小球超滤液) forms under hydrostatic pressure
- Small solutes pass freely through the filtration barrier
- Proteins are largely excluded due to size and charge
- The filtrate composition is similar to plasma but protein-free



Glomerular Ultrafiltrate Formation

- **Glomerular filtration rate (GFR)** (肾小球滤过率) averages 125 ml/min
- Total daily filtrate is approximately 180 liters
- This is equivalent to filtering the entire blood volume every 40 minutes
- About 99% of filtrate must be reabsorbed to maintain body fluid balance

Regulation of GFR

- **Sympathetic nerve** (交感神经)
stimulation affects kidney function
- Causes vasoconstriction of **afferent arterioles** (入球小动脉)
- Decreases GFR during fight-or-flight response or cardiovascular shock
- Helps preserve blood volume and redirect blood to vital organs



Regulation of GFR

- **Renal autoregulation** (肾脏自身调节) maintains stable GFR despite blood pressure changes
- Two mechanisms: **myogenic** (肌源性) response and **tubuloglomerular feedback** (球管反馈)
- The **macula densa** (致密斑) senses tubular fluid composition
- Releases ATP or adenosine to constrict afferent arterioles when needed
- This protects nephrons from overload and maintains consistent filtration

Regulation of GFR

- Summary of GFR regulation mechanisms
- Sympathetic nerves, autoregulation, and hormones control afferent arteriole diameter
- These mechanisms ensure adequate waste excretion while conserving body fluids



Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Filtration and Reabsorption Overview

- **Reabsorption** (重吸收) returns 99% of filtered fluid to the blood
- Water and solutes move from tubular fluid back into peritubular capillaries
- This process is essential to prevent dehydration
- Urine volume varies based on body needs: 0.4 to 23 liters per day



Proximal Tubule Reabsorption

- The proximal tubule reabsorbs approximately 65% of filtered salt and water
- Na^+ is actively transported out by Na^+/K^+ -ATPase pumps
- Cl^- follows passively by electrical attraction
- Water follows by **osmosis** (渗透) through aquaporin channels



Proximal Tubule Reabsorption

- Proximal tubule cells have specialized membrane properties
- **Microvilli** (微绒毛) increase apical surface area for reabsorption
- Na^+/K^+ -ATPase pumps in basolateral membrane create Na^+ gradient
- **Secondary active transport** (次级主动转运) couples Na^+ movement with glucose and amino acids



The Countercurrent Multiplier System

- The **countercurrent multiplier system** (逆流倍增系统) creates a medullary concentration gradient
- The loop of Henle has two limbs with different permeability properties
- Ascending limb actively transports NaCl out but is impermeable to water
- Descending limb is permeable to water but not to salt
- This creates progressively higher osmolality in the medulla (300 to 1200 mOsm)



The Countercurrent Multiplier System

- The thick ascending limb uses the **Na-K-2Cl cotransporter** (Na-K-2Cl 协同转运体)
- This transporter moves Na^+ , K^+ , and two Cl^- ions together
- Na^+/K^+ -ATPase maintains the driving force
- K^+ recycles back into tubular fluid, creating a positive potential
- This mechanism is the target of powerful loop diuretics



The Countercurrent Multiplier System

- Different nephron segments have distinct transport properties
- This table summarizes active and passive transport of water, ions, and urea
- Understanding these properties explains how kidneys concentrate or dilute urine



Countercurrent Exchange

- **Vasa recta** (直小血管) blood vessels preserve the medullary gradient
- Blood flows down and up parallel to the loop of Henle
- Salt and water diffuse into vasa recta going down, and out going up
- This **countercurrent exchange** (逆流交换) prevents washout of the gradient



Countercurrent Exchange

- **Urea recycling** (尿素循环) contributes significantly to medullary hypertonicity
- Urea diffuses from inner collecting duct into interstitial fluid
- Can enter the ascending limb and recirculate
- This recycling amplifies the concentration gradient needed for water reabsorption



Countercurrent Exchange

- The complete countercurrent system creates osmolality gradients throughout the kidney
- Cortex remains isotonic (300 mOsm)
- Inner medulla becomes highly hypertonic (up to 1200 mOsm)
- This gradient is essential for producing concentrated urine



ADH and Water Reabsorption

- **Antidiuretic hormone (ADH)** (抗利尿激素) regulates water reabsorption in collecting ducts
- ADH causes insertion of **aquaporin-2** (AQP2) channels into apical membrane
- Without ADH, channels remain in intracellular vesicles
- With ADH, vesicles fuse with membrane, allowing water passage



ADH and Water Reabsorption

- ADH secretion responds to plasma osmolality and blood volume
- **Osmoreceptors** (渗透压感受器) in hypothalamus detect dehydration
- Increased ADH causes water retention and concentrated urine
- Decreased ADH causes water excretion and dilute urine
- This negative feedback maintains plasma osmolality homeostasis



ADH and Water Reabsorption

- Summary of ADH regulation and actions
- Stimuli for release include increased osmolality and decreased blood volume
- ADH acts through aquaporin-2 insertion mechanism
- **Diabetes insipidus** (尿崩症) results from ADH deficiency or resistance



Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Secretion and Clearance

- **Secretion** (分泌) is the opposite of reabsorption
- Substances move from peritubular capillaries into tubular fluid
- Active transport mechanisms remove drugs, toxins, and metabolic wastes
- Both reabsorption and secretion affect renal plasma clearance



Secretion and Clearance

- Three processes affect renal clearance of substances
- **Filtration** alone: clearance equals GFR
- **Reabsorption**: decreases clearance below GFR
- **Secretion**: increases clearance above GFR
- Understanding these relationships is essential for pharmacology



Measuring GFR with Inulin

- **Inulin** (菊粉) is the gold standard for measuring GFR
- Freely filtered at the glomerulus
- Not reabsorbed or secreted by tubules
- All filtered inulin appears in urine
- Inulin clearance precisely equals GFR



Measuring GFR with Inulin

- **Creatinine** (肌酐) clearance is used clinically to estimate GFR
- Creatinine is produced by muscle metabolism
- Mostly filtered, slightly secreted
- Elevated plasma creatinine indicates reduced kidney function
- This is a key test for **renal insufficiency** (肾功能不全)

Measuring Renal Blood Flow with PAH

- **Para-aminohippuric acid (PAH)** (对氨基马尿酸) measures renal blood flow
- PAH is both filtered and secreted by **organic anion transporters** (有机阴离子转运体)
- Nearly 100% of PAH is removed in one pass through kidneys
- PAH clearance approximates total renal plasma flow (625 ml/min)



Measuring Renal Blood Flow with PAH

- Comparison of how different substances are handled by nephrons
- Shows filtration, reabsorption, and secretion patterns
- Includes clinically important molecules like glucose, urea, and creatinine



Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Potassium Balance

- **Potassium** (鉀) handling involves both reabsorption and secretion
- Almost all filtered K^+ is reabsorbed in proximal tubule
- All urinary K^+ comes from secretion in cortical collecting duct
- This allows independent regulation of K^+ balance



The Juxtaglomerular Apparatus

- The **juxtaglomerular apparatus** (肾小球旁器) regulates blood pressure and volume
- **Granular cells** (颗粒细胞) in afferent arteriole secrete **renin** (肾素)
- **Macula densa** (致密斑) cells sense tubular NaCl concentration
- This structure is key to the renin-angiotensin-aldosterone system



The Juxtaglomerular Apparatus

- Factors regulating renin and aldosterone secretion
- Decreased blood pressure and low Na^+ stimulate renin release
- Renin triggers the RAAS cascade
- **Aldosterone** (醛固酮) increases Na^+ reabsorption and K^+ secretion



The Renin-Angiotensin-Aldosterone System

- The RAAS maintains Na^+ balance and blood pressure
- Low Na^+ intake triggers renin secretion
- Renin converts **angiotensinogen** (血管紧张素原) to **angiotensin I** (血管紧张素 I)
- ACE converts angiotensin I to **angiotensin II** (血管紧张素 II)
- Angiotensin II stimulates aldosterone and causes vasoconstriction
- Negative feedback completes the regulatory loop



Ion Interactions

- Na^+ , K^+ , and H^+ secretion are interrelated
- Na^+ reabsorption creates electrical driving force
- K^+ and H^+ compete for secretion
- **Hyperkalemia** (高钾血症) can lead to acidosis
- **Hypokalemia** (低钾血症) can lead to alkalosis



Acid-Base Regulation

- Kidneys regulate blood pH by controlling H^+ and HCO_3^- levels
- H^+ is secreted into tubular fluid
- HCO_3^- is reabsorbed into blood
- **Carbonic anhydrase** (碳酸酐酶) catalyzes key reactions
- Urinary buffers (phosphate and ammonia) prevent excessive acidification



Acid-Base Regulation

- Categories of acid-base disturbances and their characteristics
- **Metabolic acidosis** (代谢性酸中毒): decreased pH, decreased HCO_3^-
- **Metabolic alkalosis** (代谢性碱中毒): increased pH, increased HCO_3^-
- **Respiratory acidosis/alkalosis**: caused by CO_2 retention or loss
- Kidneys provide compensatory responses to maintain pH balance



Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Diuretics

- **Diuretics** (利尿剂) increase urine volume by inhibiting water reabsorption
- **Loop diuretics** (襻利尿剂) like furosemide are most potent
- Act on thick ascending limb, blocking Na-K-2Cl cotransporter
- **Thiazides** (噻嗪类) act on distal tubule
- **Potassium-sparing diuretics** block aldosterone or Na^+ channels
- Used to treat hypertension, edema, and heart failure



Diuretics

- Comparison of different diuretic classes
- Shows mechanisms, sites of action, and effects on electrolytes
- Loop and thiazide diuretics cause **hypokalemia** (低钾血症)
- Potassium-sparing diuretics prevent K^+ loss
- Understanding these differences guides clinical use



Kidney Disease and Testing

- Renal function tests assess kidney health
- **Blood urea nitrogen (BUN)** (血尿素氮) and **creatinine** (肌酐) indicate filtration function
- **Urinalysis** (尿液分析) detects abnormalities
- **Hematuria** (血尿): blood in urine
- **Proteinuria** (蛋白尿): indicates glomerular damage
- **Glycosuria** (糖尿): glucose in urine, seen in diabetes
- **Oliguria** (少尿): decreased output in kidney failure

Kidney Disease and Testing

- **Glomerulonephritis** (肾小球肾炎) is inflammation of glomeruli
- Can follow streptococcal infections
- Causes hematuria, proteinuria, and edema
- **Chronic kidney disease (CKD)** (慢性肾脏病) involves progressive loss of function
- **Polycystic kidney disease** (多囊肾病) is a genetic disorder
- Treatment options include **dialysis** (透析) and **transplantation** (移植)

Outline

17.1 Structure and Function of the Kidneys

17.2 Glomerular Filtration

17.3 Reabsorption of Salt and Water

17.4 Renal Plasma Clearance

17.5 Renal Control of Electrolyte and Acid-Base Balance

17.6 Clinical Applications

Summary

Summary

- The kidneys filter 180 L of blood daily, reabsorbing 99% to produce 1-2 L of urine
- Nephrons are the functional units with specialized segments for filtration, reabsorption, and secretion
- GFR is regulated by sympathetic nerves and autoregulation mechanisms
- The countercurrent multiplier system enables production of concentrated urine
- ADH regulates water reabsorption through aquaporin-2 channels
- RAAS controls Na^+ balance and blood pressure
- Kidneys maintain acid-base balance by secreting H^+ and reabsorbing HCO_3^-
- Diuretics and other drugs target specific nephron transport mechanisms
- Kidney disease assessment uses GFR, creatinine, and urinalysis